Sea lions, walrus, and seals

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Carnivora
FAMILIES	Otariidae, Odobenidae, Phocidae
SPECIES	34

Although clumsy on land, pinnipeds (seals, sea lions, and the walrus) are supremely agile underwater. They have a streamlined body and powerful flippers and can dive to depths of over 330 ft (100 m). Some species can remain underwater for over an hour. There are 3 families: the Otariidae are the eared seals (sea lions and fur seals), which have small external ears and back flippers that can be rotated forward for movement on land; the walrus (Odobenidae), which has distinctive tusks; and the true seals (Phocidae), which have no external ears and they cannot rotate the back flippers. Only eared seals and the walrus can support themselves in a semiupright position on land. Pinnipeds are found worldwide, mostly in temperate and polar seas.

Temperature control

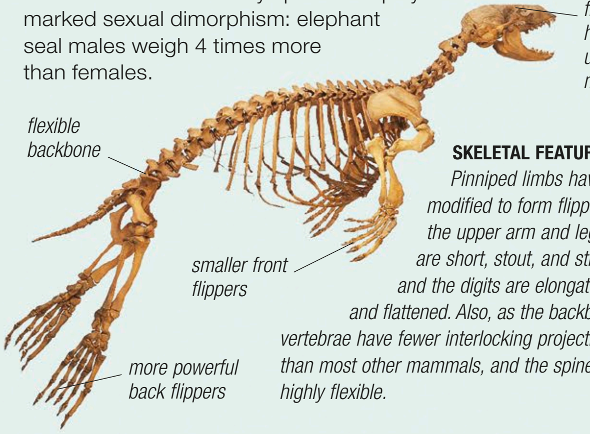
Pinnipeds have several heat-regulating adaptations. In cold water, the blubber insulates the internal organs, and blood flow to the flippers is restricted. In warm conditions, some species wave their flippers to expel excess heat. In addition, true seals and walruses can either contract the blood vessels near the skin's surface (to reduce heat loss in icy water) or they can dilate these vessels to gain heat when basking in the sun. Eared seals, however, will enter the water to avoid overheating.

COLOR CHANGE
The blood vessels in the skin of these walruses are dilated to maximize the amount of heat they can absorb by lying in the sun. As a result, their bodies turn pink.

UNDERWATER ACROBATS
In water, pinnipeds, such as these South American sea lions, are graceful, athletic, and capable of swimming at high speed. While underwater, they can communicate by sounds produced using air retained in the lungs.

Anatomy

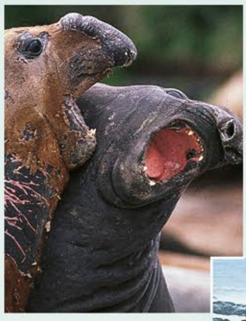
Most pinnipeds have a short face, a thick neck, and a torpedo-shaped flexible body. A layer of blubber beneath the skin provides insulation, aids buoyancy, acts as an energy reserve, and protects the organs. All species are covered with hair, except the walrus, which is nearly hairless. Pinnipeds have large eyes for good deep-water vision, excellent hearing, ear passages and nostrils that can be closed underwater, and long whiskers that enhance the sense of touch. Many species display marked sexual dimorphism: elephant seal males weigh 4 times more than females.



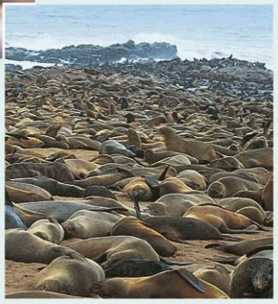
SKELETAL FEATURES
Pinniped limbs have been modified to form flippers: the upper arm and leg bones are short, stout, and strong, and the digits are elongated and flattened. Also, as the backbone's vertebrae have fewer interlocking projections than most other mammals, and the spine is highly flexible.

Life cycle

Unlike the other marine mammals (cetaceans and manatees and the dugong), pinnipeds have not abandoned land entirely. In most species, during the annual breeding season, males attempt to set up territories on suitable beaches, fighting savagely for space and excluding weaker males. Females move onto the beaches, sometimes several weeks after the males, and give birth. A few days after a pup is born (usually only one young is produced), the female mates with the male in whose territory she has settled. For the majority of the gestation period, which lasts approximately 8–15 months, pinnipeds are mostly at sea and return to land only when it is time to repeat the breeding process.



MALE AGGRESSION
In the breeding season, there is fierce competition between males, such as these 2 elephant seals, for mating rights. Only the strongest males are able to secure a breeding territory.



LARGE COLONY
Breeding beaches are always crowded. Colonies of brown fur seals number several thousand, with males controlling harems of 7–9 females.